📄 Business Intelligence Internship Report

Insurance Policy Sales & Claims Analytics

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Project: Data Simulation & Analytics Assignment
Tools Used: Python, Pandas, Power BI

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1 Project Overview

This project simulates insurance policy sales and claims data to analyze profitability and claim risk patterns. The dataset represents policy purchases made during the year 2024 and claims activity occurring in 2025 and early 2026. The objective of the analysis is to evaluate premium collection, claim trends, and loss ratios across different policy tenures.

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2 Data Simulation Approach

The policy sales dataset was generated for **1,000,000 customers** who purchased car insurance policies during 2024.

Key assumptions used:

- Each vehicle value = **₹100,000**
- Premium charged = **₹100 per year of policy tenure**
- Policy tenure distribution:
  - 20% → 1 year
  - 30% → 2 years
  - 40% → 3 years
  - 10% → 4 years
- Policy start date = **Purchase date + 365 days**

Claims simulation assumptions:

- In **2025**, 30% of vehicles purchased on the **7th, 14th, 21st, and 28th** filed claims.
- Claim amount = **10% of vehicle value**.
- In **2026**, 10% of policies with **4-year tenure** filed a second claim.

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3 Data Analysis

The following metrics were calculated using Python and visualized using Power BI:

- Total premium collected
- Total claim cost
- Claim trend by month
- Policy sales by month
- Claim ratio by policy tenure
- Policy distribution by tenure

The Power BI dashboard was created to visualize these metrics interactively.

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4 Dashboard Insights

1 Premium vs Claims

Total premium collected is significantly higher than total claims paid, indicating overall portfolio profitability.

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2 Policy Tenure Distribution

Most policies belong to the **3-year tenure category (≈40%)**, indicating customer preference for medium-term coverage.

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3 Claims Trend

Claims are concentrated in specific months due to simulated defect conditions in the dataset.

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4 Claim Ratio by Tenure

Longer tenure policies tend to have higher claim exposure, indicating increased long-term risk for the insurer.

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5 Policy Sales Pattern

Policy purchases are distributed fairly evenly across the year, suggesting stable demand for insurance policies.

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5 Conclusion

The analysis demonstrates how insurance companies can simulate historical data to evaluate profitability and risk patterns. Using analytical tools such as Python and Power BI enables businesses to monitor claim behavior, assess loss ratios, and make informed pricing decisions.

Dashboard Visualization